

Adult guide - E04 Look Again: How We Check

Version: review 1.0.0 Story: The Garden Welcome Sign Author: Maria Smith Status: working review; not an approved final publication

What this book is for

E04 is a read-aloud story and activity book for Early Years / Foundation. It continues directly from E03 at the garden gate. The child helps Mia, Sol, Tavi and one new guest, Ivo, rebuild and check a four-picture welcome sign before the morning visitors arrive.

The child should increasingly be able to:

- look at a shown source and say what is observed rather than guess;
- build, cover and rebuild a small arrangement;
- keep a failed attempt so it can be compared with the source;
- follow a short ordered trace to find the first changed step;
- understand that one measuring tool answers only its declared question;
- ask a friend to make a fresh check without copying the first answer; and
- accept that checking may confirm, disagree or show that more work is needed.

This is not a general classroom checking lesson with SFT language added. Its sequence comes from the maintained E04 syllabus and the receipt-backed SFT boundaries below.

Opening recap and invitation

Book Four and Level Four must make sense to a child who has just finished Level Three, has read only the books, or selects Level Four later. The opening should:

1. welcome the child back to Mia, Sol and Tavi;
2. recall that they repaired the turning-light trail by keeping a failed try;
3. show the trio at the exact garden gate teased at the end of E03;
4. reveal one clear new problem: the four-picture welcome sign must be rebuilt and checked before the morning visitors arrive;
5. introduce Ivo as the sole new guest, with his moss-green body, round lens and checkerboard satchel visible before his job is described;
6. show the source plan, blank sign and closed gate so the mission has a visible cause; and
7. preview looking, building, remembering, tracing, measuring and friend checking without revealing any challenge answer.

The story contains no unexplained package, ramp, magical instruction note or new location jump. Every move follows from the same sign-and-gate mission.

Scientific authority and exact boundary

SFT-FOUNDATION-DERIVATION-TRACE-001

- Claim title: Complete replayable derivation trace
- Receipt:
sha256:457414dd7e3a14e44b200b1ac329cdc9e2bb3527abe901c3255d7fb6f9233c5e
- Status: model admitted, depth-independent, independently replicated
- Exact role in E04: an admitted finite derivation has one complete ordered source- and dependency-bound record that deterministically replays every registered operation, intermediate and terminal result.

The four picture-placement cards translate the experience of keeping steps in order. They are not derivation identities, and replaying a classroom sign does not prove the formal claim.

SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001

- Claim title: One-way derivation-to-measurement boundary
- Receipt:
sha256:a4e409f6bb8b203de746bd0c21d3f2566e8052003219490c657249e11f6f2086
- Status: model admitted, depth-independent, independently replicated
- Exact role in E04: the exact source-bound law and target question are fixed before measurement is opened; measurement preserves every row and cannot select or mutate the law.

The WIDTH and HEIGHT ribbons translate the one-question boundary. A width result answers width only. It cannot establish picture order, height, beauty, truth or SFT authority. A measuring tool answers only the question we chose and declared.

SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001

- Claim title: Single fail-closed SFT admission law
- Receipt:
sha256:0e21ffcb217271ddfab4000901691761c2b7fd423b93d27f81ab757f0300c58c
- Status: model admitted, depth-independent, independently replicated
- Exact role in E04: the complete fail-closed authority path preserves adverse, rejected and unresolved dispositions; an incomplete route does not gain authority.

Sol's first rebuild, its changed step and the width-only conclusion remain in the checking book. The closed gate is an educational image of a halt: it opens only after every declared classroom checkpoint has a visible record. The gate is not the formal admission engine.

SFT-ENG-INDEPENDENT-CHECK-001

- Claim title: Implementation-distinct check
- Receipt:
sha256:927d1b0f871cb92e06ae180a9c1fbcf6b2fe0a024135fb487fbf41a9b082b364
- Status: model admitted, empirically tested and independently reconstructed
- Branch: Engineering Translation

- Exact bounded role in E04: an independent check reconstructs through a separately identified implementation path while preserving specifications, results, disagreements and provenance.

Ivo uses the original source, fresh pieces and a separate work area without seeing the team's answer. This is a child-scale translation of a fresh path. It does not claim that two children constitute formal independent certification.

What the activities do not prove

A picture plan, memory game, measuring ribbon, garden gate or friend's answer is a bounded educational demonstration. None derives, validates or changes an SFT law. The book does not assert that agreement makes a statement true, that disagreement makes it false, or that a physical fit confers scientific status.

The following remain excluded:

- measurement or observed success selecting a derivation;
- one tool answering a property it was not chosen to measure;
- self-checking relabelled as independent checking;
- deletion of wrong, missing, unresolved or inconvenient attempts;
- guessing, voting, adult confidence or praise as evidence;
- binary digits or signed proof labels;
- semantic numerical zero;
- floating-point equality;
- completed infinity or an ungenerated continuum; and
- outside models, standards or curricula as authority in SFT derivation.

Story cause-chain

Keep the mission audible: rebuild and check the Garden Welcome Sign before the morning visitors arrive.

- Pages 3-5: E03's repaired trail reaches the closed garden gate; the sign problem appears; Ivo joins as the sole new guest.
- Pages 6-8: the child observes one requested source picture before meeting the observation/guess distinction.
- Pages 9-11: the child builds all four pictures while the source is visible.
- Pages 12-14: the source is covered; Sol rebuilds from memory; the source is uncovered; the mismatching attempt stays visible.
- Pages 15-16: the child compares every marked corner and identifies the two changed pictures.
- Pages 17-19: the child follows the placement trace and finds Step 3 as the first departure from the source.
- Pages 20-22: the repaired sign's width is measured; the result is explicitly prevented from answering picture or height questions.

- Pages 23-25: Ivo reconstructs separately and confirms all four picture places.
- Pages 26-27: Mia overextends the width result; Ivo disagrees; a separate height check resolves the named missing question.
- Pages 28-30: every checkpoint question is matched to its own record before the gate opens.
- Pages 31-32: the narrator addresses the child directly; the checked sign guides visitors; a closed Star Door quietly teases later learning without inventing an unsupported title.

If the child loses the story, point to the source sign, rebuilt sign and closed gate. Restate the mission rather than skipping to an answer.

Challenge and reveal answers

- Page 7 - Which picture is at MOON? The source shows BEE. Ask the child to point to the actual source picture before naming it.
- Page 10 - Build from the source. Sunflower goes to SUN, bee to MOON, watering can to LEAF and boot to STAR. Use every card once.
- Page 13 - Does Sol's memory rebuild match? It does not. Honest answers include NOT SURE while the source is covered. Page 14 reveals that bee and boot traded places.
- Page 15 - Which pictures differ? BEE and BOOT. Compare all four corners; do not infer that every picture changed.
- Page 18 - Where did the swap begin? STEP 3 first departs from the source by placing BOOT at MOON. Step 4 is also different, but it is later.
- Page 21 - Does the width fit? FITS. Both sign edges align with the green WIDTH marks.
- Page 24 - Do the two picture checks agree? CONFIRM. Ivo's separately rebuilt picture positions match the repaired team sign.
- Page 26 - What should happen when the checkers disagree? CHECK HEIGHT. Guessing or voting cannot answer the missing height question.
- Page 29 - Can every checkpoint be supported? Yes: source comparison supports picture positions, the width record supports width, the height record supports height and Ivo's fresh board supports the friend check.

Wrong answers are useful records. Keep the child's selected card or route long enough to ask, "Which question are we answering?" and "Which record supports that answer?" Avoid "No, that is wrong," "You are clever," or revealing the correct object with a pointing gesture before the attempt.

What the narrator explains after each game

Each explanation comes after a completed attempt. It names the action, result, idea and story purpose without giving away the next stage.

1. Look-and-Point: You moved the lens over the shown plan and found the bee in the moon corner. That was an observation because you used what you could see instead of guessing. A clear observation gives the team a starting record.
2. Build-the-Board: You used every picture once and matched each one to its source corner. Building from a shown source lets the team compare the result with the same

starting plan.

3. **Curtain Memory:** You studied, covered and rebuilt the four pictures. When the curtain opened, the first rebuild stayed visible even though two pictures had traded places. A failed try can still help us check.
4. **Difference Finder:** You compared every corner and found that bee and boot changed places while the other two pictures stayed the same. Comparing the complete source with the complete try is more reliable than one quick guess.
5. **Footprint Trace:** You followed the placement record in order and found Step 3 as the first change. A trace keeps what happened first, next and last, so the team can find where a different result began.
6. **Measuring Ribbon:** You aligned the WIDTH ribbon and found that the sign fits between the hooks. The ribbon answered width only; it did not check the pictures or height. A measuring tool answers its declared question.
7. **Friend Check:** Ivo rebuilt from the same source without seeing the team sign. His four picture places matched, so the fresh check confirmed that result. A separate path matters because copying would not be a fresh check.
8. **Disagreement Detective:** Mia used the width answer to claim the whole sign fit, but Ivo noticed height was still unchecked. You chose a height tool instead of guessing or voting. Disagreement can show that more work is needed.
9. **Final Checkpoint:** You matched every gate question to its own visible record. The sign passed only after picture, width, height and friend checks were supported. Keeping all the records made the final decision clear.

Canonical final fourth-wall explanation

The game's spoken line and visible caption must match this text exactly:

You learned how to check. You looked at what was shown instead of guessing.
You built the sign, covered it, rebuilt it, and kept the first try when it did
not match. You followed the steps to find where the change began. You used a
width tool for width and a height tool for height, because one tool answers
only its own question. Then Ivo checked by a fresh path. His result confirmed
the picture places. When Mia and Ivo disagreed, you did more work instead of
guessing or voting. Checking matters because it helps us say what we
observed, keep useful mistakes, and find the next question that needs an
answer.

Plain-language meanings

- **Observation:** saying what we can see in the shown source or scene.
- **Guess:** choosing an answer without checking what is shown.
- **Source:** the kept starting plan used for a build and its checks.
- **Rebuild:** make the complete arrangement again from its source.
- **Failed attempt:** a try that did not meet the declared check. Keep it.

- Trace: a kept record of the steps in the order they happened.
- Compare: look at two complete records for sameness and difference.
- Width: distance from one side to the other side.
- Height: distance from the bottom to the top.
- Independent check: a friend uses the same source by a fresh path without copying the first answer.
- Confirm: the fresh check agrees with the first result for the named question.
- Disagree: two results or claims do not match.
- More work: the current records do not yet answer every declared question.

Do not use deterministic replay, target opening, disposition receipt, implementation carrier, provenance class or fail-closed admission as unexplained child shorthand.

Running the book activity

Use four large picture cards, a sheet divided into four marked spaces and a cloth. Draw the source before the activity. The child's pictures may be simple but each must have a word or shape cue so colour is never the only identifier.

1. Read pages 3-8 and point to what the source actually shows.
2. Build the sign with the source visible.
3. Photograph or sketch the first build if it cannot safely remain in place.
4. Cover the source and rebuild on a second sheet.
5. Uncover and compare every marked space. Keep the second build even if it differs.
6. Make a four-card trace of the child's placement order.
7. Ask a width question and a height question separately with separate strips of paper.
8. If possible, let a second person rebuild from the source without seeing the first answer. If that is not possible, describe the idea; do not call a second look by the same person independent.

The child may point, speak, use eye gaze or choose between large cards. Physical rebuilding is optional.

Safe physical activity

Use large cards that cannot be swallowed, a light paper or cardboard board and a soft cloth. Do not hang a heavy sign, use a working gate, give a child a glass lens, or ask them to stretch a rigid or retracting metal tape. The game's lens is fictional; a real activity should use a cardboard viewing ring. Adult hands operate scissors, fasteners and elevated displays.

Accessibility

- Every picture has a name above it and a distinct shape; colour is secondary.
- The four spaces use sun, moon, leaf and star markers instead of position or colour alone.
- WIDTH and HEIGHT tools carry both words and directional edge cues.

- Source, first rebuild, repaired sign and Ivo's check use persistent labels.
- Challenge choices remain visible until the separate reveal.
- All descriptions name the relevant source, attempt, step, tool and result.
- The accessible HTML preserves page order and exposes each picture description beside the page text.
- Every game drag or movement has a labelled tap and keyboard alternative.
- Reduced-motion mode removes decorative movement without removing state.
- Local narration must match captions exactly. Sound and colour are never required to understand an answer.

Companion Level Four contract

Level Four retains Mia, Sol and Tavi as the permanent trio and introduces only Ivo. It opens with a magical recap and invitation, then presents nine connected stages. Each stage has a genuinely different replayable mechanic:

1. sweep a magnifying lens over a picture plan and tag the requested observed picture while decoys move between rounds;
2. place four picture pieces into four symbol-marked sign spaces, with shuffled layouts that are always solvable and a clear reset;
3. study a varied board, close the curtain, rebuild it and uncover the same source, retaining the exact attempt;
4. compare two full boards and mark all changed positions before the mistake allowance runs out;
5. guide Sol along ordered placement footprints and stop at the first trace step that departs from the source;
6. align a tactile width ribbon between two hooks, with too-short, fit and too-wide rounds that answer width only;
7. direct Ivo through a separate build route, then connect matching results without exposing the team's answer during his build;
8. choose the missing question and operate a height tool while guess and vote choices visibly fail to answer it; and
9. route source, width, height and friend-check evidence cards to four gate checkpoints; unsupported or mismatched evidence halts the gate and can be retried.

Every stage includes meaningful success, recoverable mistakes, feedback, retry, replay and bounded variation. No stage is a nested sequence of “Next” buttons. Characters move within each scene, and the active tool or record is large enough for phone touch.

Level Four has its own gentle garden-checking score, distinct from Levels One through Three. It may use soft wooden taps, leaf rustles and short lens chimes, but music must never signal an answer. Music, narration and effects have separate remembered controls. App switching pauses all audio; returning resumes the current visual state without replaying an old or completed line.

The level stores stage, round, source board, exact failed attempt, trace, measurement questions and completed checks. Returning to the menu and resuming must restore those

exact records. Completion provides replay, level select and the sequential next-learning option when such a reviewed level exists.

Connecting the book and Level Four

The book and game teach the same caused story in distinct ways. The book gives time for challenge-and-reveal comparison. The game lets the child operate each tool, make a recoverable mistake and replay with a new bounded arrangement.

- Begin with: “We are rebuilding and checking one welcome sign before the visitors arrive.”
- Ask “What can you see?” before “What do you think?”
- Keep the source and first try in view after a reveal.
- When stuck, restate the named question and point to the available records; do not point to the correct choice.
- After a game explanation, invite the child to find the same action in the book. Neither medium is homework for the other.

Optional picture codes

Codes never reveal an answer, skip an attempt or grant learning progress.

- LOOKCLOSE - page 8: Ivo's lens reflects the four corner symbols.
- SIGNMAKER - page 11: the completed frame grows a harmless flower vine.
- KEEPFIRST - page 14: Sol adds a “useful try” ribbon to the saved board.
- SPOTCHANGE - page 16: bee and boot perform a cosmetic place-swap replay.
- FOLLOWSTEPS - page 19: the four trace cards make four soft leaf sounds.
- WIDTHONLY - page 22: the width ribbon rolls itself into Ivo's satchel.
- FRIENDCHECK - page 25: matching picture pairs exchange a glow.
- MOREWORK - page 27: Mia and Ivo share a friendly checking wave.
- ALLCHECKED - page 30: the open gate plays a short garden greeting.

External references

The UK Early Years Foundation Stage framework may calibrate language length, safe activity, communication and accessibility. Publisher conventions and other educational games may inform usability comparisons. None is authority for SFT claims, topic order or derivation.

Review and publication status

Review version 1.0.0 remains under edu/. It may be rendered and tested as a live review work. It must not be copied into publications/education, described as finally approved or released as a child-facing hosted service until Maria Smith explicitly approves this exact book and companion level.